Capstone Project: Full Alert-to-Response Cycle

1. Objective

The objective of this project is to simulate a real-world cyber attack, detect it using a SIEM system (Wazuh), perform triage, execute response actions, and document the entire incident lifecycle from detection to mitigation.

2. Tools Used

- Metasploit Framework
- Metasploitable2 (Target: **172.23.134.132**)
- iptables *(*used as an alternative to CrowdSec due to installation limitation in metasploitable *)*

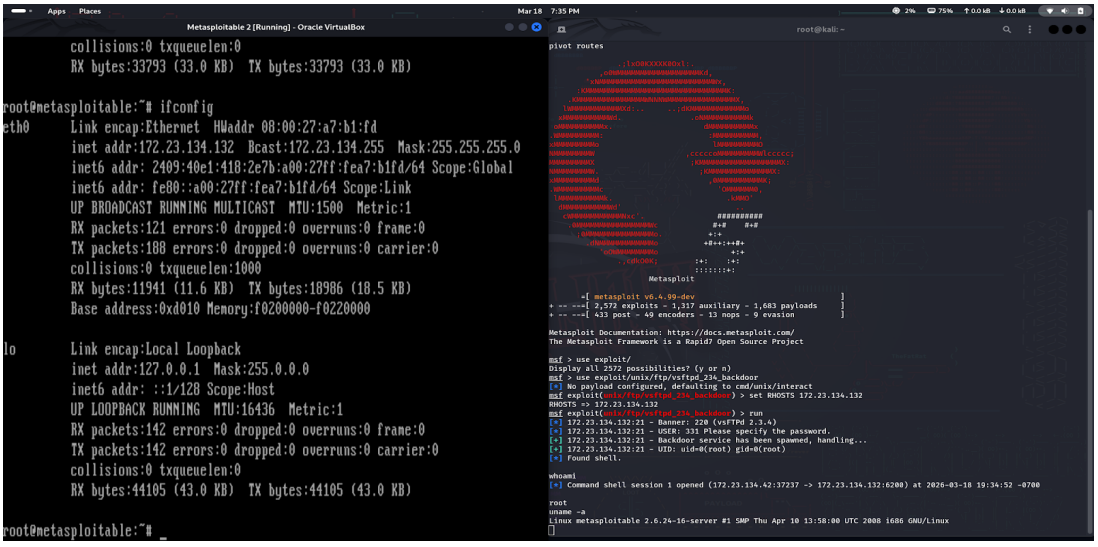
3. Attack Simulation

A vulnerability in the Metasploitable2 machine was exploited using Metasploit.

Exploit Details

- Module: `exploit/unix/ftp/vsftpd_234_backdoor`
- Attacker: Kali Linux
- Target: 172.23.134.132(Metasploitable VM)

The exploit successfully triggered a backdoor, allowing unauthorized shell access to the target system.



4. Detection and Triage

Wazuh SIEM was used to monitor system logs and detect suspicious activity related to the attack.

Alert Log Table

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-17 11:00:00	172.23.134.42	VSFTPD Exploit Attempt	T1190

Analysis

- The alert indicated exploitation of a vulnerable FTP service
- Activity matched **Initial Access technique (MITRE ATT&CK T1190)**
- The behavior was consistent with exploitation attempts using known backdoor vulnerabilities

5. Response

The attacker IP was identified from logs and blocked to prevent further access.

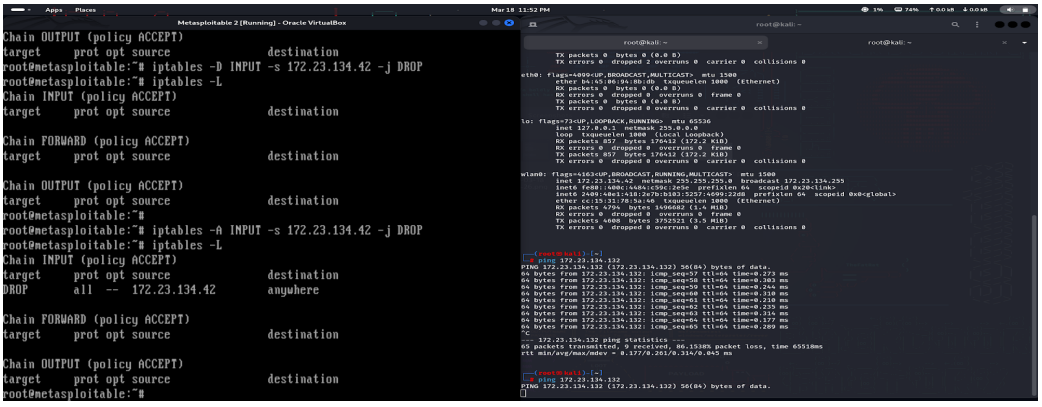
Action Taken

Since CrowdSec could not be installed due to technical limitations, iptables was used:

```
iptables -A INPUT -s 172.23.134.42 -j DROP
```

Verification

- After blocking the IP, connectivity from the attacker machine was denied
- Ping/connection attempts failed, confirming successful containment



6. Incident Report

Executive Summary

A controlled attack was conducted using Metasploit to exploit a known vulnerability (vsftpd backdoor) in a Metasploitable2 system. The attacker successfully gained unauthorized access, simulating a real-world compromise scenario. The activity was monitored and analyzed using Wazuh SIEM, which provided visibility into suspicious behavior. Immediate response actions were taken to mitigate the threat.

Timeline

- Attack initiated from Kali Linux
- Exploit executed using Metasploit module
- Backdoor shell access obtained
- Suspicious activity identified in logs
- Attacker IP detected
- IP blocked using firewall rules

Recommendations

- Disable vulnerable and outdated services (e.g., insecure FTP versions)
- Apply regular security patches
- Deploy automated threat prevention tools such as CrowdSec
- Improve log monitoring and alerting mechanisms
- Implement network segmentation and access control

7. Stakeholder Briefing

A simulated cyber attack was performed to evaluate the organization's security monitoring and response capabilities. The attacker exploited a known vulnerability in a test system and gained unauthorized access. The activity was monitored and analyzed using Wazuh. Immediate action was taken to block the attacker's IP address using firewall rules, preventing further access. This was conducted in a controlled lab environment with no impact on real systems. The exercise highlights the importance of proactive monitoring, rapid response, and proper security configurations. Future improvements include automated threat blocking and regular vulnerability patching.